

Small-Signal Stability Analysis and MATLAB Simulation of a Single-Machine Infinite-Bus System

Based on Kundur, *Power System Stability and Control*, Chapter 12

Power System Stability — Chapter 12 Study

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Abstract

The single-machine infinite-bus (SMIB) system—a synchronous generator connected through an equivalent reactance to an ideal constant-voltage, constant-frequency bus—is the canonical model for studying rotor-angle small-signal stability. *Small-signal stability* is the ability of the machine to remain in synchronism following a small disturbance; because the disturbance is small, the system equations may be linearized about an operating point and stability judged from the eigenvalues of the resulting state matrix. This report analyzes the SMIB system at four levels of detail following Kundur, *Power System Stability and Control*, Chapter 12: the *classical* model (constant flux behind transient reactance), the *field-circuit* model (adding field-flux dynamics), the *AVR* model (adding a thyristor exciter and automatic voltage regulator), and the *AVR + PSS* model (adding a power system stabilizer). For each model the analysis builds the linearized state-space representation, computes the eigenvalues, and evaluates the damping ratio and oscillation frequency, with time-domain responses used to confirm the eigenvalue predictions. The principal findings reproduce the classical electromechanical swing frequency of 1.02 Hz; show that field-circuit dynamics contribute a small positive damping; demonstrate that a high-gain AVR, in the presence of a negative K_5 , *reduces* damping torque and can drive the swing mode unstable (eigenvalue $+0.504 \pm j7.23$); and confirm that a properly tuned PSS restores positive damping and returns the swing mode to the stable left-half plane ($-1.005 \pm j6.607$, $\zeta = 0.15$). All results agree with the corresponding Kundur Chapter 12 examples.

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1 Introduction

Small-signal stability is the ability of a power system to maintain synchronism when subjected to small disturbances such as incremental changes in load or generation. Because such disturbances are small, the nonlinear system equations may be linearized about a steady-state operating point. Stability is then governed by the eigenvalues of the linearized state matrix: the system is small-signal stable if and only if every eigenvalue has a negative real part.

The dominant concern for a synchronous generator connected to a large grid is the *electromechanical* or *swing* mode—an oscillation of the rotor against the rest of the system, typically in the range 0.1–2 Hz. The character of this mode is set by two torque components acting on the rotor: a *synchronizing* torque (in phase with the rotor-angle deviation $\Delta\delta$) and a *damping* torque (in phase with the speed deviation $\Delta\omega_r$). Insufficient synchronizing torque produces a non-oscillatory (monotonic) instability; insufficient damping torque produces a growing oscillation.

This report studies the four canonical SMIB models of Kundur Chapter 12. For each, the linearized state matrix is assembled, its eigenstructure is computed, and the result is interpreted physically and validated against the textbook. The analysis is implemented and reproduced numerically in MATLAB; the implementation is summarized in Section 7, but the emphasis of the report is the stability analysis itself.

2 Objectives

1. Present the SMIB system and the meaning of small-signal stability for rotor-angle dynamics.
2. Develop the linearized state-space model of the generator at four levels of detail: classical, field-circuit, AVR, and AVR + PSS.
3. Derive the Heffron–Phillips K -constants from the fundamental d – q machine parameters, including magnetic saturation.
4. Evaluate stability through eigenvalues, damping ratios, and natural frequencies, and confirm the conclusions with time-domain responses.
5. Demonstrate the effect of the AVR and the corrective action of the PSS on the swing mode, and validate every result against Kundur Chapter 12.

3 The Single-Machine Infinite-Bus System

The single-machine infinite-bus (SMIB) system represents one synchronous generator connected, through a transformer and transmission line of combined equivalent reactance, to a very large power system. The large system is idealized as an *infinite bus*: a source of constant voltage magnitude, constant frequency, and fixed phase angle that is unaffected by the behaviour of the single machine. The infinite bus therefore serves as the angular and frequency reference against which the generator’s rotor swings.

In the classical representation the generator is modelled as a constant internal voltage $E' \angle \delta$ behind the transient reactance X'_d , and the external network as a series reactance

X_E . The rotor angle δ is the angle of the internal voltage relative to the infinite-bus voltage $E_B \angle 0^\circ$, as shown in Figure 1.

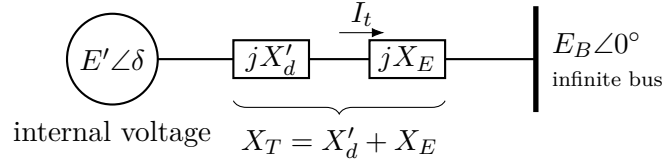


Figure 1: The single-machine infinite-bus system. The generator is represented by a constant voltage E' behind the transient reactance X'_d , connected through the external reactance X_E to an infinite bus of constant voltage $E_B \angle 0^\circ$. The rotor angle δ is measured relative to the infinite-bus reference; the total reactance is $X_T = X'_d + X_E$.

The SMIB system is used as the standard model for rotor-angle small-signal stability because it isolates the electromechanical interaction between one machine and the grid while remaining low-order enough to be analyzed in closed form. It is the starting point for the linearized machine models developed in the standard references on power system dynamics [1, 2, 3]. The physical mechanisms it exposes—synchronizing and damping torque, the destabilizing effect of high-gain voltage regulation, and the corrective role of a power system stabilizer—carry over directly to machines embedded in large multi-machine systems.

4 Mathematical Modelling

4.1 Power-Angle Relationship

With the generator represented as $E' \angle \delta$ behind X'_d and the network as X_E , the air-gap (electrical) power delivered to the infinite bus is

$$P_e = \frac{E'E_B}{X_T} \sin \delta, \quad X_T = X'_d + X_E. \quad (1)$$

This is the power-angle curve: power transfer increases with rotor angle up to $\delta = 90^\circ$. The slope of this curve at the operating point determines how strongly the machine resists a change in angle—the synchronizing torque.

4.2 Swing Equation

The rotor motion is governed by the swing equation, which balances the accelerating torque (mechanical input minus electrical output) against the rotor inertia:

$$\frac{2H}{\omega_0} \frac{d^2 \delta}{dt^2} = T_m - T_e - K_D \Delta \omega_r, \quad (2)$$

where H is the inertia constant, $\omega_0 = 2\pi f_0$ the base electrical speed, K_D a damping coefficient, and $\Delta \omega_r$ the per-unit speed deviation. The mechanical and electrical torques are nearly equal to the corresponding powers in per unit near synchronous speed.

4.3 Linearization

For small deviations about the operating point δ_0 , the electrical power of Eq. (1) is linearized as

$$\Delta P_e = \left. \frac{\partial P_e}{\partial \delta} \right|_{\delta_0} \Delta \delta = \underbrace{\frac{E' E_B}{X_T} \cos \delta_0}_{K_S} \Delta \delta = K_S \Delta \delta, \quad (3)$$

where K_S is the *synchronizing torque coefficient*. Substituting the linearized power into the swing equation and writing the rotor angle and speed deviations as states gives the state-space model below.

4.4 State-Space Model: Classical

With the generator represented by a constant voltage E' behind X'_d , the linearized swing equations are (Kundur Eq. 12.78):

$$\frac{d}{dt} \begin{bmatrix} \Delta \omega_r \\ \Delta \delta \end{bmatrix} = \begin{bmatrix} -\frac{K_D}{2H} & -\frac{K_S}{2H} \\ \omega_0 & 0 \end{bmatrix} \begin{bmatrix} \Delta \omega_r \\ \Delta \delta \end{bmatrix} + \begin{bmatrix} \frac{1}{2H} \\ 0 \end{bmatrix} \Delta T_m. \quad (4)$$

The undamped natural frequency and damping ratio follow from the characteristic equation of Eq. (4):

$$\omega_n = \sqrt{\frac{K_S \omega_0}{2H}}, \quad \zeta = \frac{1}{2} \frac{K_D}{2H \omega_n}. \quad (5)$$

4.5 State-Space Model: Field-Circuit Dynamics

Adding the field-winding flux linkage $\Delta \psi_{fd}$ as a third state captures the slow demagnetizing effect of armature reaction. The state matrix becomes (Kundur Eq. 12.116):

$$A_B = \begin{bmatrix} -\frac{K_D}{2H} & -\frac{K_1}{2H} & -\frac{K_2}{2H} \\ \omega_0 & 0 & 0 \\ 0 & a_{32} & a_{33} \end{bmatrix}, \quad (6)$$

where K_1 is the synchronizing torque coefficient at constant flux, K_2 relates electrical torque to field flux, and the field-equation entries are $a_{32} = -K_3 K_4 / T_3$, $a_{33} = -1/T_3$. The constants K_1 – K_4 and the effective field time constant T_3 are derived from the d – q parameters (Section 6).

4.6 State-Space Model: Exciter and AVR

A thyristor exciter (Kundur type ST1A) with gain K_A and a terminal-voltage transducer of time constant T_R adds a fourth state Δv_1 . Two further constants are required: K_5 (the change in terminal voltage with rotor angle) and K_6 (the change in terminal voltage with field flux). The field and transducer rows are:

$$a_{34} = -\frac{K_3}{T_3} K_A, \quad \Delta \dot{v}_1 = \frac{K_5}{T_R} \Delta \delta + \frac{K_6}{T_R} \Delta \psi_{fd} - \frac{1}{T_R} \Delta v_1. \quad (7)$$

The field-flux response to a rotor-angle perturbation, including exciter action, is (Kundur Eq. 12.144 with $G_{ex}(s) = K_A$):

$$\frac{\Delta\psi_{fd}}{\Delta\delta}(s) = \frac{-K_3[K_4(1 + sT_R) + K_5K_A]}{(1 + sT_3)(1 + sT_R) + K_3K_6K_A}. \quad (8)$$

Evaluating $K_2 \cdot \Delta\psi_{fd}/\Delta\delta$ at $s = j\omega$ (the swing frequency) decomposes the field/AVR torque into a synchronizing part (real) and a damping part (imaginary, scaled by ω_0/ω). With $K_5 < 0$, increasing K_A raises synchronizing torque but reduces damping torque—the central trade-off of high-gain AVRs.

4.7 State-Space Model: Power System Stabilizer

The PSS restores damping by injecting a supplementary signal derived from $\Delta\omega_r$ through a washout and lead-lag compensator (Kundur Eq. 12.151):

$$G_{PSS}(s) = K_{STAB} \frac{sT_w}{1 + sT_w} \frac{1 + sT_1}{1 + sT_2}. \quad (9)$$

The washout state Δv_2 and lead-lag output Δv_s extend the model to six states. The stabilizer output enters the exciter summing junction, adding phase lead that converts the AVR's negative damping back to positive.

4.8 Summary of the Four Models

The four models form a hierarchy of increasing detail. Each adds states that capture an additional physical effect, as summarized in Table 1.

Table 1: State-space model summary. Each model augments the previous one with states representing an additional physical subsystem.

Model	State vector	Purpose
Classical	$[\Delta\omega_r, \Delta\delta]^T$	Basic electromechanical swing mode
Field-circuit	$[\Delta\omega_r, \Delta\delta, \Delta\psi_{fd}]^T$	Adds field-flux (rotor circuit) dynamics
AVR	$[\Delta\omega_r, \Delta\delta, \Delta\psi_{fd}, \Delta v_1]^T$	Adds exciter / voltage-regulator effect
AVR + PSS	$[\Delta\omega_r, \Delta\delta, \Delta\psi_{fd}, \Delta v_1, \Delta v_2, \Delta v_s]^T$	Adds supplementary damping through the stabilizer

4.9 Eigen-Analysis

For state matrix A with eigenvalue $\lambda = \sigma \pm j\omega$:

$$\zeta = \frac{-\sigma}{\sqrt{\sigma^2 + \omega^2}}, \quad f = \frac{|\omega|}{2\pi} \text{ Hz}. \quad (10)$$

The right eigenvector gives the *mode shape* (relative activity and phase of each state in the mode); the participation factor $p_{ki} = |v_{ki}| |w_{ik}|$ measures the contribution of state k to mode i , dimensionless and independent of units.

5 Small-Signal Stability Analysis Procedure

The stability of each model is evaluated by the following systematic procedure, applied identically to all four model levels:

1. **Operating point.** Solve the steady-state network and machine equations for the rotor angle δ_0 , the internal voltage E' , and the d - q currents and fluxes.
2. **K -constants.** Derive (or, where the textbook supplies them, adopt) the Heffron–Phillips constants K_1 – K_6 and the field time constant T_3 from the operating point and machine parameters.
3. **State matrix.** Assemble the linearized state matrix A for the chosen model level.
4. **Eigenvalues.** Compute the eigenvalues $\lambda_i = \sigma_i \pm j\omega_i$ of A .
5. **Stability test.** Determine stability from the real parts: the system is stable if and only if $\sigma_i < 0$ for all i .
6. **Damping and frequency.** For each oscillatory mode compute the damping ratio ζ_i and frequency f_i .
7. **Time-domain validation.** Confirm the eigenvalue prediction with the time response (rotor angle, speed, and electrical power) to a small disturbance.

6 Derivation of the K -Constants from d - q Parameters

The K -constants of the field-circuit model are computed from the fundamental machine parameters rather than supplied directly. The procedure follows Kundur Example 12.3:

1. **Saturation.** The total air-gap flux ψ_{at} at the operating point sets the saturation factors. Above the threshold ψ_{T1} , the incremental saturated inductance uses $K_{sd}^{incr} = 1/(1 + B_{sat}\psi_I)$ with $\psi_I = A_{sat} \exp[B_{sat}(\psi_{at} - \psi_{T1})]$.
2. **Operating point.** The internal angle is obtained directly as $\delta_0 = \text{atan2}(E_{Bd0}, E_{Bq0})$, from which the d - and q -axis currents i_{d0}, i_{q0} , the field current i_{fd0} , and the field voltage $E_{fd0} = L_{adu} i_{fd0}$ follow.
3. **Network-coupling coefficients.** The constants m_1, n_1, m_2, n_2 (Kundur Eqs. 12.106–12.108) couple the rotor-angle and field-flux perturbations to the d - q currents. A subtlety confirmed during validation: m_2, n_2 use the *incremental* L_{ads} (not the parallel L'_{ads}).
4. **K -constants.** K_1, K_2 are assembled from m_1, n_1, m_2, n_2 and the operating-point fluxes; K_4 satisfies $a_{32} = -(\omega_0 R_{fd}/L_{adu})K_4$ (note L_{adu} , not L_{fd}); $T_3 = -1/a_{33}$.

6.1 Physical Meaning of the K -Constants

The six Heffron–Phillips constants each quantify one linearized coupling in the machine. Their physical meanings are summarized in Table 2.

Table 2: Physical meaning of the Heffron–Phillips K -constants.

Constant	Physical meaning
K_1	Change in electrical torque for a change in rotor angle at constant flux (synchronizing torque)
K_2	Change in electrical torque for a change in field flux linkage
K_3	Field-circuit gain factor (impedance factor of the field loop)
K_4	Demagnetizing effect of a change in rotor angle on the field flux
K_5	Change in terminal voltage for a change in rotor angle
K_6	Change in terminal voltage for a change in field flux linkage

The constant K_5 is pivotal. Under heavy load and high external reactance K_5 becomes *negative*, meaning a rotor-angle increase *lowers* the terminal voltage. A high-gain AVR then responds with a field-voltage action that is out of phase with the rotor swing, producing negative damping torque. This single sign change explains why a regulator intended to improve voltage control can degrade rotor-angle damping (Section 8.3) and motivates the PSS (Section 8.4).

Table 6 confirms the derived values against Example 12.3.

7 MATLAB Implementation

The analysis is carried out by a small set of MATLAB routines: case definitions hold the machine, network, and operating data together with the textbook golden references; an operating-point and K -constant layer derives the linearization; a state-matrix builder assembles models A–D; and an eigen-analysis routine returns eigenvalues, damping ratios, mode shapes, and participation factors. A standalone runner executes all four models end-to-end and produces the figures of Section 9. Table 3 lists the components.

Table 3: MATLAB components used to produce the results in this report.

Component	File	Role
Cases	+cases/case_kundur_smib_*.m	parameters + golden refs
Classical init	+smib/smib_classical_init.m	δ_0 , E' , K_S
d - q init	+smib/smib_dq_init.m	operating point, saturation
K -constants	+smib/smib_k_constants.m	derive K_1 – K_4 , T_3
Torque decomp.	+smib/smib_torque_components.m	K_S , K_D vs K_A
State matrix	+smib/smib_build_state_matrix.m	models A/B/C/D
Eigen-analysis	+smib/smib_analyze.m	λ , ζ , mode shape
Plots	internal/plotting/smib_plot_*.m	figures
Runner	run_smib_example.m	end-to-end analysis

8 Worked Examples and Validation

This section analyzes four worked cases from Kundur Chapter 12. For each, the problem data are stated first, then the analysis procedure of Section 5 is applied, and finally the computed result is compared against the textbook value.

8.1 Example 12.2 — Classical Model

Problem (Kundur Ex. 12.2, pp. 732–737). A 2220 MVA, 24 kV, 60 Hz generator with inertia $H = 3.5$ and transient reactance $X'_d = 0.30$ pu delivers $P = 0.9$ pu, $Q = 0.3$ pu (overexcited) at a terminal voltage $E_t = 1.0\angle 36^\circ$ pu. The external reactance to the infinite bus is $X_E = 0.65$ pu and the infinite-bus voltage is $E_B = 0.995\angle 0^\circ$ pu. Find the operating point, the synchronizing torque coefficient K_S , and the swing-mode eigenvalues for $K_D = 0, 10, -10$.

Solution

1. Terminal current from $S = E_t I_t^*$:

$$I_t = \left(\frac{P + jQ}{E_t} \right)^* = 0.9487\angle 17.57^\circ = 0.9045 + j0.2863 \text{ pu.}$$

2. Internal voltage behind X'_d (armature resistance neglected):

$$E' = E_t + jX'_d I_t = 1.1229\angle 49.91^\circ \text{ pu.}$$

3. Rotor angle relative to the infinite bus: $\delta_0 = \angle E' - \angle E_B = 49.91^\circ$.
4. Total reactance $X_T = X'_d + X_E = 0.30 + 0.65 = 0.95$ pu.
5. Synchronizing coefficient (Eq. 3):

$$K_S = \frac{|E'| |E_B|}{X_T} \cos \delta_0 = \frac{1.1229 \times 0.995}{0.95} \cos 49.91^\circ = 0.7574.$$

6. Undamped natural frequency (Eq. 5) with $\omega_0 = 2\pi(60) = 376.99$ rad/s:

$$\omega_n = \sqrt{\frac{K_S \omega_0}{2H}} = \sqrt{\frac{0.7574 \times 376.99}{7}} = 6.387 \text{ rad/s} = 1.0165 \text{ Hz.}$$

Table 4: Example 12.2 — computed vs. textbook golden values.

Quantity	Computed	Golden
δ_0 (deg)	49.91	49.92
$ E' $ (pu)	1.123	1.123
X_T (pu)	0.950	0.950
K_S	0.7574	0.757
f_n (Hz)	1.0165	1.0165
λ ($K_D=0$)	$\pm j6.387$	$\pm j6.39$
λ ($K_D=10$)	$-0.714 \pm j6.347$	$-0.714 \pm j6.35$
ζ ($K_D=10$)	0.112	0.112
λ ($K_D=-10$)	$+0.714 \pm j6.347$	$+0.714 \pm j6.36$

With $K_D = 0$ the mode is undamped (purely imaginary eigenvalues, $f = 1.02$ Hz); positive damping moves it into the left-half plane, and negative damping into the right-half plane (instability). The root locus as K_D is swept is shown in Figure 2; the rotor-angle and speed responses for $K_D = 10$ in Figure 3; and the corresponding electrical-power response in Figure 4.

8.2 Example 12.3 — Field-Circuit Dynamics

Problem (Kundur Ex. 12.3, pp. 753–757). The same machine and operating point as Example 12.2, but now modelled with field-circuit dynamics and magnetic saturation. The fundamental machine parameters (pu, machine base) are listed in Table 5. Derive the Heffron–Phillips constants K_1 – K_4 and T_3 from these parameters, assemble the 3×3 state matrix, and find the eigenvalues.

Table 5: Example 12.3 — given machine and network data.

Parameter	Value	Parameter	Value
L_{adu}	1.65	R_{fd}	0.0006
L_{aqu}	1.60	A_{sat}	0.031
L_l	0.16	B_{sat}	6.93
R_a	0.003	ψ_{T1}	0.80
L_{fd}	0.153	X_E	0.65
H	3.5	K_D	0

Solution

1. *Saturation.* Air-gap flux $\psi_{at} = |E_t + (R_a + jL_l)I_t| = 1.0604$. Since $\psi_{at} > \psi_{T1}$, $\psi_I = A_{sat}e^{B_{sat}(\psi_{at}-\psi_{T1})} = 0.1884$, giving the total and incremental saturation factors $K_{sd} = 0.8491$, $K_{sd}^{incr} = 0.4337$. Hence the saturated mutual inductances $L_{ads} = 0.8491(1.65) = 1.4011$, $L_{aqs} = 1.3586$, and the incremental $L_{ads}^{incr} = 0.7156$.
2. *Internal angle and currents.* $\delta_i = \angle[E_t + (R_a + jX_{qs})I_t] = 43.13^\circ$, from which $i_{d0} = 0.8342$, $i_{q0} = 0.4518$, $e_{d0} = 0.6836$, $e_{q0} = 0.7299$.
3. *Rotor angle and field.* Resolving onto the infinite-bus frame gives $E_{Bd0} = 0.9773$, $E_{Bq0} = 0.1876$, so $\delta_0 = \text{atan2}(E_{Bd0}, E_{Bq0}) = 79.13^\circ$. The field current $i_{fd0} = 1.4514$ yields $E_{fd0} = L_{adu}i_{fd0} = 2.395$.
4. *Coupling coefficients* (Eqs. 12.106–12.108): $m_1 = 1.0436$, $n_1 = 0.1268$, $m_2 = 0.8801$, $n_2 = 0.00176$.
5. *K-constants:* $K_1 = 0.7643$, $K_2 = 0.8649$, $K_3 = 0.323$, $K_4 = 1.4187$, and $T_3 = -1/a_{33} = 2.356$ s.

The resulting state matrix and its eigenvalues are:

$$A_B = \begin{bmatrix} 0 & -0.1092 & -0.1236 \\ 376.99 & 0 & 0 \\ 0 & -0.1945 & -0.4244 \end{bmatrix}, \quad \lambda = \begin{cases} -0.110 \pm j6.411 & \text{(swing)} \\ -0.205 & \text{(field)} \end{cases} \quad (11)$$

Table 6: Example 12.3 — derived constants and eigenvalues vs. golden.

Quantity	Derived	Golden
δ_0 (deg)	79.13	79.13
E_{fd0} (pu)	2.395	2.395
m_1	1.0436	1.0437
n_1	0.1268	0.1268
m_2	0.8801	0.8802
K_1	0.7643	0.7643
K_2	0.8649	0.8649
K_4	1.4187	1.4187
T_3 (s)	2.356	2.365
swing λ	$-0.110 \pm j6.411$	$-0.110 \pm j6.41$
field λ	-0.205	-0.204

The field circuit introduces a small positive damping (swing-mode $\zeta \approx 0.017$) and a non-oscillatory field mode near -0.204 . The participation factors confirm the physical identity of each mode: the swing mode is 49% $\Delta\omega_r$ + 49% $\Delta\delta$ with only 1.7% $\Delta\psi_{fd}$, whereas the field mode is 99.8% $\Delta\psi_{fd}$ (Table 7, visualized in Figure 5).

Table 7: Example 12.3 — participation-factor matrix (state $\times$ mode).

State	Swing mode ($-0.110 \pm j6.41$)	Field mode (-0.205)
$\Delta\omega_r$	0.492	0.001
$\Delta\delta$	0.492	0.001
$\Delta\psi_{fd}$	0.017	0.998

8.3 Table 12.1 — Effect of the AVR

Problem (Kundur Sec. 12.4, pp. 762–765). For a machine with $K_1 = 1.591$, $K_2 = 1.5$, $K_3 = 0.333$, $K_4 = 1.8$, $K_5 = -0.12$, $K_6 = 0.3$, $T_3 = 1.91$ s, $H = 3.0$, and a thyristor exciter of transducer time constant $T_R = 0.02$ s, evaluate how the synchronizing and damping torque components contributed by the field/AVR loop vary as the exciter gain K_A is increased.

Method. The field-flux response to a rotor-angle perturbation is given by Eq. 8. The contribution to electrical torque is $\Delta T_e = K_2 \Delta\psi_{fd}$. Evaluating $K_2 \cdot \Delta\psi_{fd}/\Delta\delta$ at $s = j\omega$ (with $\omega \approx 10$ rad/s, the swing frequency) and separating real and imaginary parts gives the synchronizing component $K_S(\Delta\psi_{fd})$ and the damping component $K_D(\Delta\psi_{fd}) = \text{Im}(\cdot)\omega_0/\omega$.

Table 8: Table 12.1 — torque components vs. AVR gain K_A ($\omega = 10$ rad/s).

K_A	$K_S(\Delta\psi_{fd})$	golden	$K_D(\Delta\psi_{fd})$	golden
0	-0.0025	-0.0025	+1.770	+1.772
15	-0.0093	-0.0093	+0.024	+0.024
25	-0.0098	-0.0098	-1.165	-1.166
200	+0.2801	+0.2804	-12.271	-12.272

With $K_5 < 0$ the damping coefficient crosses zero near $K_A \approx 15$ and becomes strongly negative at high gain: the AVR improves synchronizing torque at the expense of damping (Figure 6). Using the Example 12.6 K -set, a full eigen-analysis of the 4-state AVR model at $K_A = 200$ confirms an unstable swing mode at $+0.504 \pm j7.23$.

8.4 Example 12.6 — AVR plus PSS

Problem (Kundur Ex. 12.6, pp. 776–778). For the Example 12.3 machine with the high-gain AVR ($K_A = 200$) that destabilizes the rotor mode, design a power system stabilizer with $K_{STAB} = 9.5$, washout $T_w = 1.4$ s, and lead-lag $T_1 = 0.154$ s, $T_2 = 0.033$ s. Show that the PSS restores stability.

Result. The full 6-state eigenvalues with the PSS in service are

$$\lambda = \{ -39.10, -1.005 \pm j6.607, -0.739, -19.80 \pm j12.82 \},$$

all in the left-half plane. The swing mode moves from $+0.504 \pm j7.23$ (AVR only, $\zeta = -0.07$) to $-1.005 \pm j6.607$ ($\zeta = +0.15$); the migration on the complex plane is shown in Figure 7.

Table 9: Example 12.6 — swing mode with and without PSS at $K_A = 200$.

Configuration	swing λ	golden	ζ	stable
AVR only	$+0.504 \pm j7.232$	$+0.504 \pm j7.23$	-0.070	no
AVR + PSS	$-1.005 \pm j6.607$	$-1.005 \pm j6.607$	$+0.150$	yes

Adding the PSS converts the unstable AVR-only swing mode into a well-damped mode ($\zeta = 0.15$), matching Example 12.6 exactly. The well-damped exciter mode ($-19.8 \pm j12.8$, $\zeta = 0.84$) and the washout-related real mode (-0.739) also agree with the textbook.

9 Figures and Interpretation

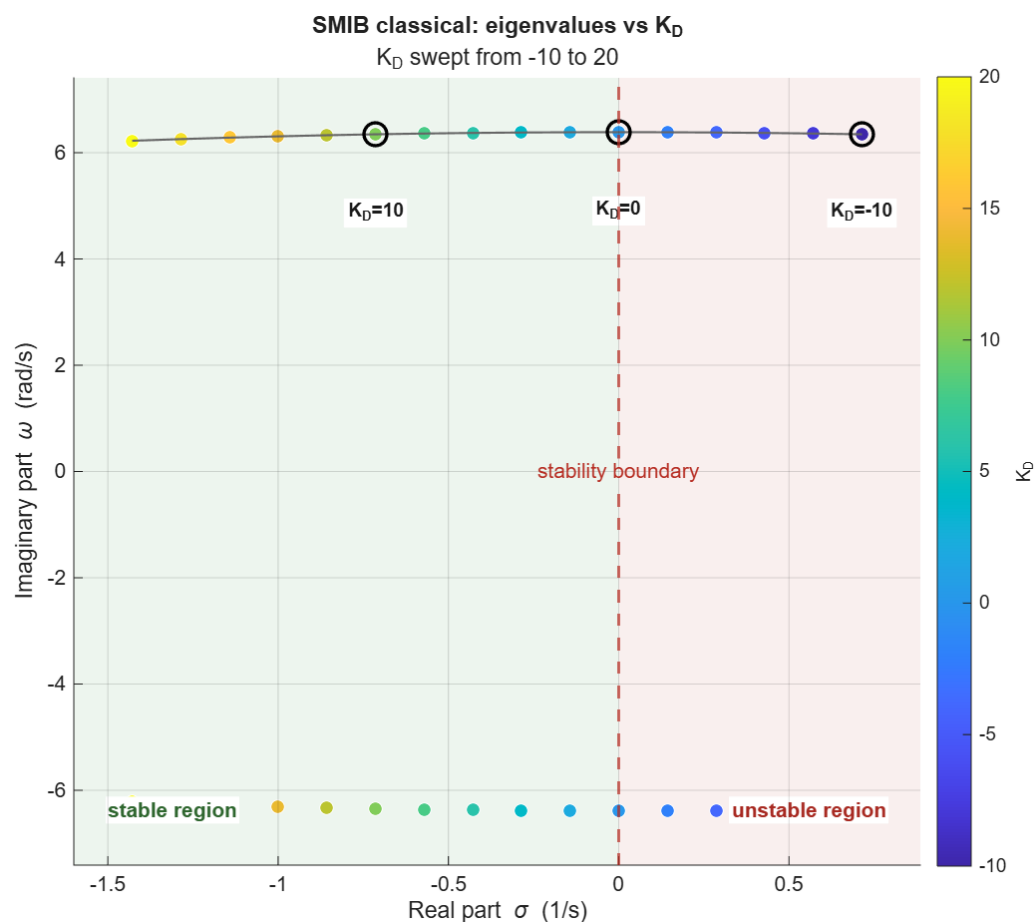


Figure 2: Eigenvalue plot / root locus of the classical SMIB model as the damping coefficient K_D is swept from -10 to $+20$. The shaded left-half plane is the stable region and the right-half plane the unstable region; the dashed imaginary axis is the stability boundary ($\sigma = 0$). Increasing positive damping K_D shifts the swing eigenvalues leftward into the stable region, while negative damping shifts them rightward into the unstable region; the marked points correspond to $K_D = -10$, 0 , and $+10$.

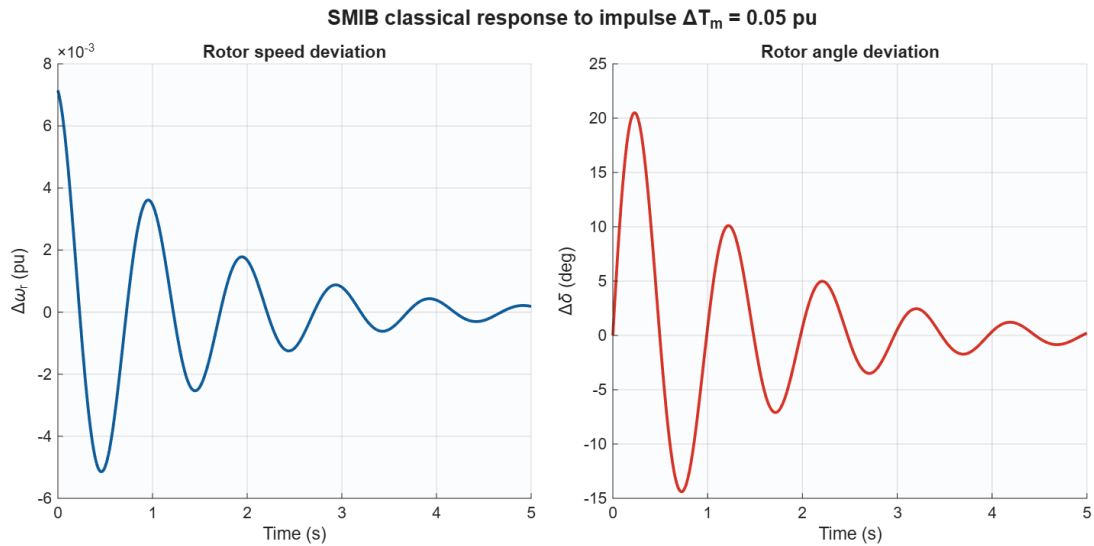


Figure 3: Time-domain response of the classical model ($K_D = 10$) to a small torque disturbance. Left: rotor speed deviation $\Delta\omega_r$, which decays to zero as the rotor returns to synchronous speed. Right: rotor angle deviation $\Delta\delta$, whose decaying ~ 1 Hz oscillation ($\zeta = 0.112$) shows that the generator remains in synchronism with the infinite bus.

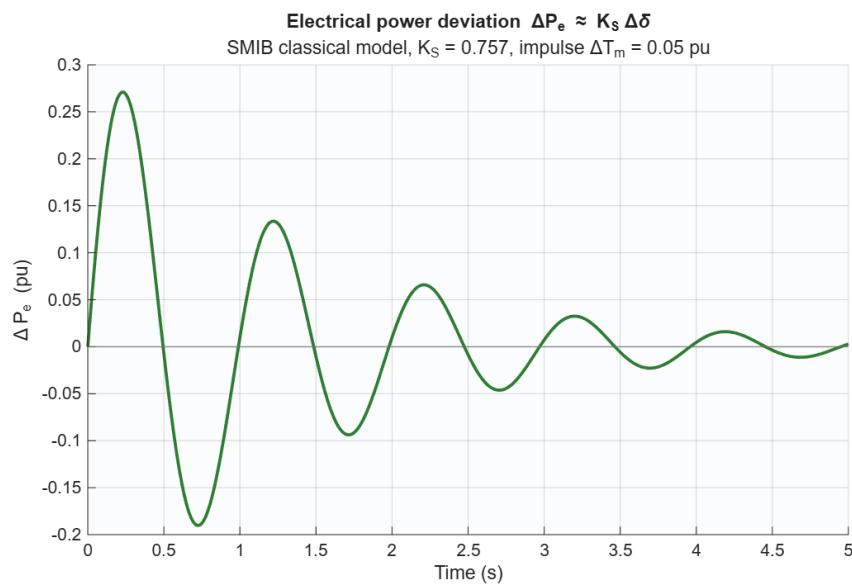


Figure 4: Electrical power response of the classical model ($K_D = 10$) to the same disturbance, computed as $\Delta P_e \approx K_S \Delta\delta$. The decaying power oscillation confirms that the machine settles back to steady power transfer with the infinite bus—the time-domain signature of a small-signal stable system.

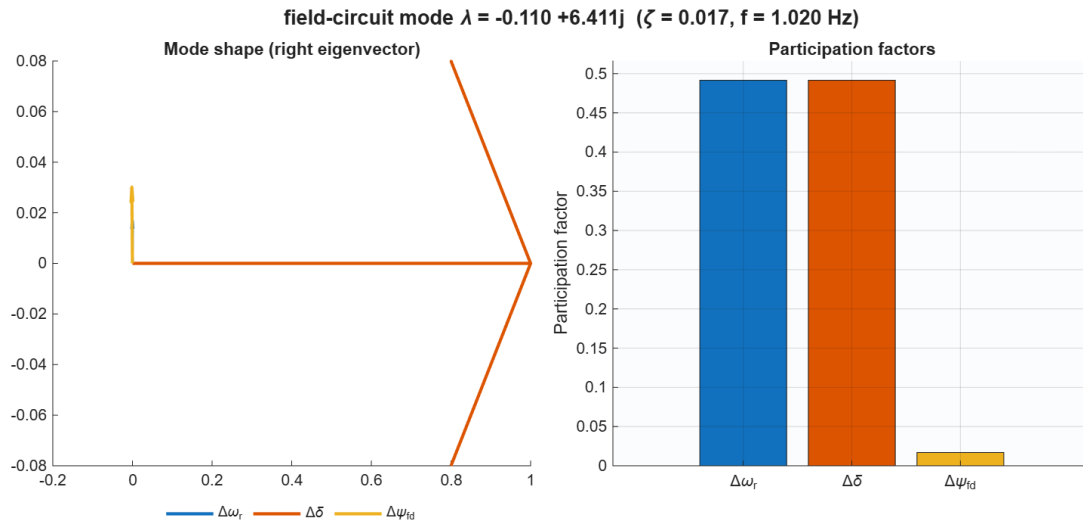


Figure 5: Mode shape (left) and participation factors (right) of the field-circuit swing mode. The rotor states $\Delta\omega_r$ and $\Delta\delta$ dominate the electromechanical mode; the field flux $\Delta\psi_{fd}$ contributes negligibly, confirming the modal identification of Table 7.

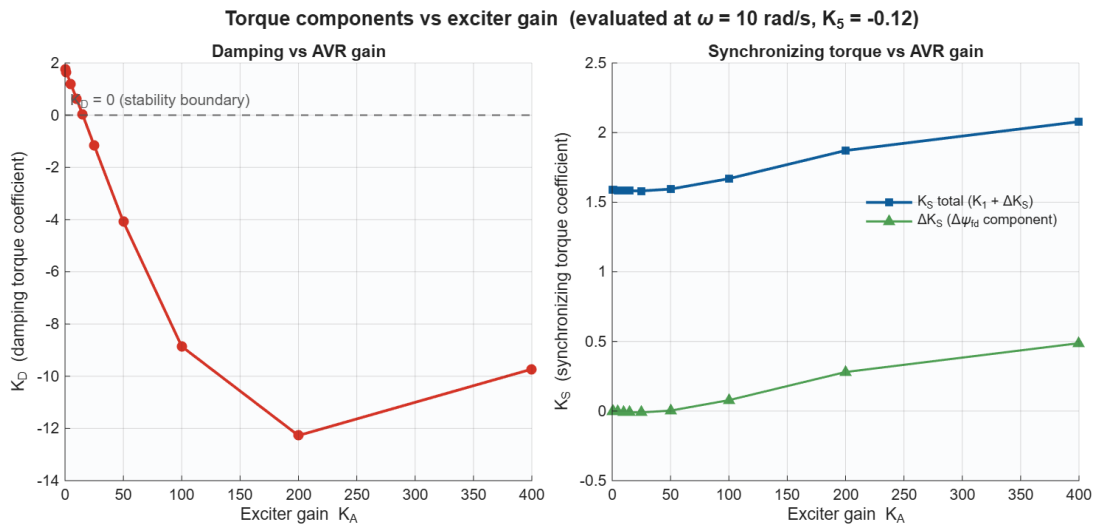


Figure 6: Damping (left) and synchronizing (right) torque coefficients versus AVR gain K_A . With $K_5 < 0$, raising the gain increases the synchronizing torque K_S but drives the damping torque K_D negative, reproducing Kundur Table 12.1. Negative damping is the mechanism by which a high-gain AVR destabilizes the swing mode.

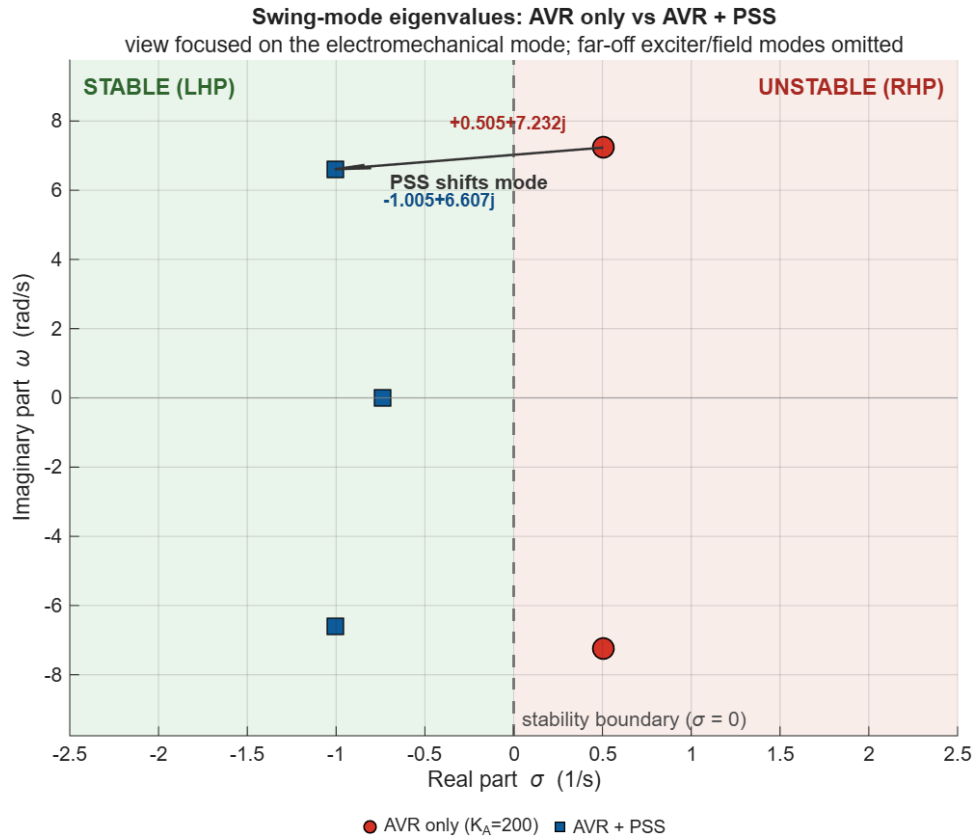


Figure 7: Swing-mode eigenvalues with AVR only (right-half plane, $+0.504 \pm j7.23$, unstable) and with AVR + PSS (left-half plane, $-1.005 \pm j6.607$, stable) at $K_A = 200$. The PSS moves the electromechanical mode across the stability boundary into the stable region. The view is focused on the swing mode; the far-off exciter and field modes lie outside the frame.

Table 10 summarizes how each figure is read in stability terms.

Table 10: Interpretation of the figures in small-signal stability terms.

Figure	What it shows	Stability interpretation
Eigenvalue plot / root locus (Fig. 2)	Movement of eigenvalues as K_D varies	Left-half plane $\Rightarrow$ stable; right-half plane $\Rightarrow$ unstable
Rotor angle response (Fig. 3, right)	Oscillation of $\Delta\delta$	Decaying oscillation $\Rightarrow$ synchronism is maintained
Rotor speed response (Fig. 3, left)	Oscillation of $\Delta\omega_r$	Return to zero $\Rightarrow$ speed returns to synchronous
Electrical power response (Fig. 4)	Oscillation of ΔP_e	Decaying power oscillation $\Rightarrow$ stable power transfer
AVR gain effect (Fig. 6)	K_S and K_D versus K_A	High gain raises K_S but can reduce K_D
AVR vs PSS eigenvalues (Fig. 7)	Swing mode before and after PSS	PSS moves the mode to the stable left-half plane

10 Discussion

The four models together tell a single, coherent story about rotor-angle small-signal stability.

- **Classical model.** The classical model fixes the basic electromechanical swing frequency near 1 Hz, set by the synchronizing coefficient K_S and the inertia H (Eq. 5). With no damping ($K_D = 0$) the mode sits exactly on the imaginary axis.
- **Role of K_D .** Positive K_D moves the swing eigenvalues into the left-half plane (damped, stable); negative K_D moves them into the right-half plane (growing oscillation, unstable). The root locus of Figure 2 traces this directly.
- **Field-circuit dynamics.** Adding the field flux introduces a small *positive* damping of the swing mode ($\zeta \approx 0.017$) and a separate, well-damped non-oscillatory field mode. The electromechanical mode remains clearly identifiable through its participation factors.
- **High-gain AVR.** A thyristor AVR sharply improves voltage regulation and synchronizing torque, but because $K_5 < 0$ at this loading it injects *negative* damping torque. At $K_A = 200$ the damping coefficient reaches -12.3 and the swing mode crosses into the right-half plane ($+0.504 \pm j7.23$): the regulator has destabilized the rotor mode.
- **Power system stabilizer.** The PSS adds a supplementary signal derived from speed deviation, phase-compensated by a washout and lead-lag network, to the exciter summing junction. This restores positive damping and returns the swing mode to the left-half plane ($-1.005 \pm j6.607$, $\zeta = 0.15$), as Figure 7 shows.

The progression illustrates the fundamental trade-off in excitation control: the same high gain that strengthens synchronizing torque and voltage regulation can erode damping, and a dedicated stabilizer is the standard remedy.

11 Conclusion

The small-signal stability of the single-machine infinite-bus system has been analyzed at four levels of detail using a linearized state-space formulation and eigenvalue analysis, with time-domain responses confirming the eigenvalue predictions throughout. The state matrix of each model was assembled from the machine and network parameters—with the Heffron–Phillips K -constants derived from first-principles d - q data and saturation—and its eigenstructure was used to judge stability and to quantify the damping and frequency of the electromechanical mode.

The analysis establishes the central conclusions of rotor-angle stability: state-space and eigenvalue analysis evaluate SMIB small-signal stability directly through the real parts of the eigenvalues; the time-domain rotor-angle, speed, and electrical-power responses corroborate those predictions; a high-gain AVR improves voltage regulation and synchronizing torque but, when $K_5 < 0$, can reduce damping and destabilize the swing mode; and a properly tuned power system stabilizer restores positive damping and moves the unstable swing mode back into the stable left-half plane. Every numerical result—the 1.02 Hz

classical swing frequency, the field-circuit eigenvalues, the AVR damping-torque trend of Table 12.1, and the PSS stabilization from $+0.504 \pm j7.23$ to $-1.005 \pm j6.607$ —agrees with the corresponding examples of Kundur Chapter 12.

References

- [1] P. Kundur, *Power System Stability and Control*, McGraw-Hill, 1994. Chapter 12 (Examples 12.2, 12.3, 12.6; Section 12.4, Table 12.1).
- [2] P. M. Anderson and A. A. Fouad, *Power System Control and Stability*, 2nd ed., IEEE Press / Wiley, 2003.
- [3] P. W. Sauer and M. A. Pai, *Power System Dynamics and Stability*, Prentice Hall, 1998.